

Solon Falas, Ph.D.

Machine Learning Engineer | Data Scientist | Physics-Informed & Robust Deep Learning | Nicosia, Cyprus

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Professional Summary

ML Engineer and Data Scientist (Ph.D., Computer Engineering) with 7+ years of applied research, including 5+ years building deep-learning models. Specialise in **physics-informed neural networks** and **adversarially robust ML** for safety-critical cyber-physical systems. Own the full ML lifecycle — data pipelines, feature engineering, training (**PyTorch**, **TensorFlow**), hyperparameter optimisation (Optuna), and evaluation — with first-author results in *IEEE Transactions on Industrial Informatics*.

Technical Skills

Programming: Python, Bash/Shell, C/C++/C#, LaTeX

ML Methods: Deep learning, physics-informed neural networks (PINNs), regression & state estimation, adversarial robustness, model evaluation & benchmarking

ML Frameworks: PyTorch, TensorFlow, scikit-learn, Optuna

Data Science: pandas, NumPy, Matplotlib, Seaborn, Jupyter

Tools: Linux, Git, MATLAB, PowerWorld

AI Tooling: Claude, Codex, GitHub Copilot, LLM-assisted R&D, agentic workflows, spec/test-driven development, prompt engineering

Domains: Cyber-physical systems, smart grid & power systems, IoT & embedded security

Education

Ph.D., Computer Engineering (2017–2025)

University of Cyprus, Nicosia, Cyprus

Cross-Layer Physics-based Security for Cyber-Physical Systems

B.Sc., Computer Engineering (2013–2017)

University of Cyprus, Nicosia, Cyprus

Awards

- **Best Computer Engineering Undergraduate Dissertation**, University of Cyprus (2017)
- **Ph.D. Scholarship**, KIOS Research and Innovation Center of Excellence (2017–2025)

Hobbies & Activities

- Personal finance, portfolio construction, risk analysis, investing
- Running, weightlifting, fitness, sports
- Digital privacy, DIY self-hosting, smart-home projects

Languages

Greek (Native) | English (Fluent)

Work Experience

Research Associate – Machine Learning & Security (2025–Present)

KIOS Research and Innovation Center of Excellence, University of Cyprus

- Conduct applied ML research and engineering on robust, physics-informed models for cyber-physical systems and embedded security.

Researcher – Machine Learning & Data Science (2017–2025)

KIOS Research and Innovation Center of Excellence, University of Cyprus

- Built and evaluated physics-informed neural networks (PINNs) for power-system state estimation under noisy, faulty, and adversarial operating conditions, embedding AC power-flow constraints as physics-based regularisation to improve generalisation; published in *IEEE TII* (2025).
- Built end-to-end Python data pipelines to ingest, preprocess, and curate multi-scenario simulation datasets; ran systematic hyperparameter optimisation with Optuna and analysed estimation-error metrics across diverse operating conditions.
- Maintained reproducible ML workflows with version-controlled code, documented configurations, and benchmarking reports using Python, TensorFlow, PyTorch, and MATLAB.
- Designed an end-to-end secure firmware-update framework for embedded/IoT devices (*ACM JETC*).

Teaching Assistant – Computer Engineering (2017–2025)
University of Cyprus

- Delivered lab instruction and technical support for digital logic, microprocessors, operating systems, and VLSI CAD courses.

Selected ML Projects

Robust Power-System State Estimation with Physics-Informed Neural Networks

Built TensorFlow/PyTorch PINN models with AC power-flow regularisation; ran Optuna hyperparameter sweeps and benchmark evaluation across fault and adversarial-measurement scenarios; established reproducible training and reporting workflow.

Learning Without Adversarial Training: A PINN for Secure Power System State Estimation under False Data Injection Attacks

Developed a physics-informed neural network that stays robust to false data injection attacks without adversarial training, using embedded power-flow constraints to resist manipulated measurements while preserving accuracy under normal operating conditions.

Selected Publications

- S. Falas et al., “Robust Power System State Estimation Using Physics-Informed Neural Networks,” *IEEE Transactions on Industrial Informatics*, 2025.
- S. Falas et al., “Data Manipulation Attack Mitigation in Power Systems Using Physics-Informed Neural Networks,” *IEEE Int. Conf. on Cyber Security and Resilience (CSR)*, 2025.
- S. Falas et al., “A Modular End-to-End Framework for Secure Firmware Updates on Embedded Systems,” *ACM J. on Emerging Technologies in Computing Systems*, 2022.